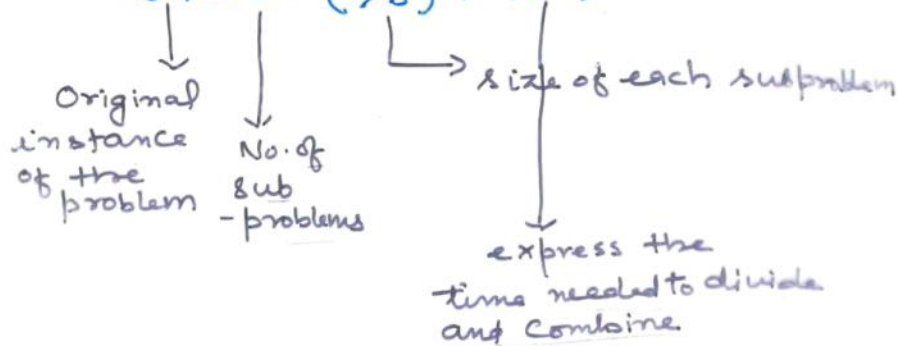


Divide and Conquer (Contd....) (Solution of Recurrences)



$$T(n) = aT(n/b) + f(n)$$



{ The Substitution Method }:-

- Guess the form of the solution
 - * by Experience and creativity
 - * By some heuristics
- Use mathematical induction to find the constants and show that the solution works.

Example:- $T(n) = 2T(\lfloor n/2 \rfloor) + n$

— Guess the solution: $T(n) = O(n \log n)$

Need to prove :- $T(n) \leq cn \log n$ for some c

— prove the soln by induction

$$T(n) \leq 2(c \lfloor n/2 \rfloor \log \lfloor n/2 \rfloor) + n$$

$$\leq cn \log(n/2) + n$$

$$\leq cn \log n - cn + n$$

$$\leq cn \log n \quad (c \geq 1)$$

$$\therefore T(n) \leq cn \log n$$

Hypothesis

$$T(k) \leq ck \log k$$

Example -:

②

* Find the bound:

$$T(n) = 2T(n/2) + n \quad (O(n \log n))$$

* How about $T(n) = 2T(\lfloor n/2 \rfloor) + n$?

* How about $T(n) = 2T(\lceil n/2 \rceil) + n$?

$$- T(n) \leq 2T(n/2 + 1) + n$$

- Domain Transformation

$$\bullet \text{ set } S(n) = T(n+a)$$

$$\text{assume } S(n) = 2S(n/2) + n \therefore S(n) = O(n \log n)$$

$$\bullet S(n) = 2S(n/2) + n \rightarrow T(n+a) = 2T(n/2 + a) + n$$

$$\therefore T(n) \leq 2T(n/2 + 1) + n \rightarrow T(n+a) \leq 2T((n+a)/2 + 1) + n$$

$$\therefore \text{set } \frac{n}{2} + a = \frac{(n+a)}{2} + 1 \therefore a = 2$$

$$\therefore T(n) = S(n-2) = O((n-2) \log(n-2)) \\ = O(n \log n)$$

- As a result, ceiling, floor, and lower terms will not affect

$$T(n) = 2T(n/2 + 1) + n ?$$

(3)

- {changing variables} -

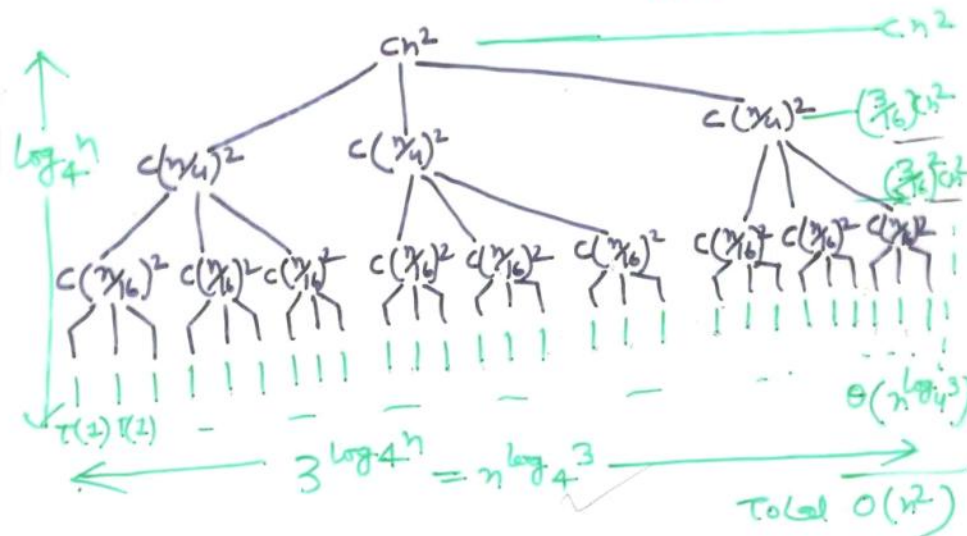
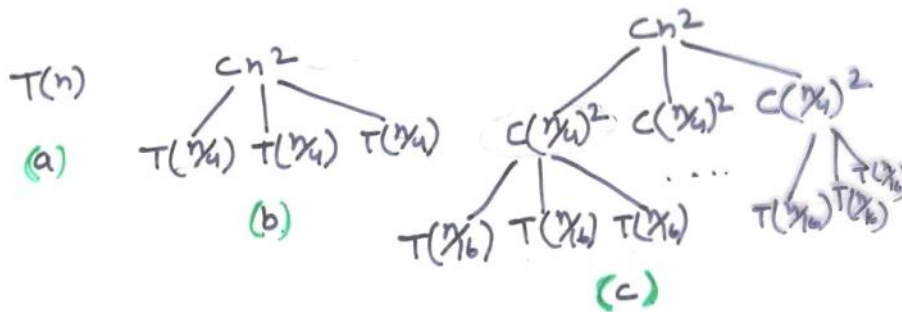
- * suppose $T(n) = 2T(\sqrt{n}) + \lg n$
- * Rename $m = \lg n$
 $\therefore T(2^m) = 2T(2^{m/2}) + m$
- * Domain Transformation
- $S(m) = T(2^m)$ so $S(m) = 2S(m/2) + m$
- solution
 $S(m) = O(m \lg m)$
 $T(n) = T(2^m) = S(m) = O(m \lg m)$
 $= O(\lg n \lg \lg n)$

The recurrence - Tree Method -

- * {Ideas} -
 - Each node represents the cost of a single subproblem
 - Sum up the costs with each level to get level cost
 - Sum up all the level costs to get Total cost
- * Suitable for divide and conquer recurrence

④

— Recursion tree for
 $T(n) = 3T(\lfloor n/4 \rfloor) + \Theta(n^2)$



(5)

- {Master Method} -

— for $T(n) = aT(n/b) + f(n)$,
[n/b may be $\lceil n/b \rceil$ or $\lfloor n/b \rfloor$]
— $a \geq 1$, $b > 1$, $f(n)$ be a function.

1. if $f(n) = O(n^{\log_b a - \epsilon})$ for some $\epsilon > 0$
Then $T(n) = \Theta(n^{\log_b a})$
2. if $f(n) = \Theta(n^{\log_b a})$ then
 $T(n) = \Theta(n^{\log_b a} \lg n)$
3. if $f(n) = \Omega(n^{\log_b a + \epsilon})$ for some $\epsilon > 0$
and if $a f(n/b) \leq c f(n)$
for some $c < 1$ and all sufficiently
large n , then $T(n) = \Theta(f(n))$

Implications of Master Theorem -

* Comparison between $f(n)$ and $n^{\log_b a}$
($<, =, >$)

* Must be asymptotically smaller (or larger) by a polynomial i.e. n^ϵ for some $\epsilon > 0$

* In case 3, the 'regularity' must be satisfied $a f(n/b) \leq c f(n)$ for some $c < 1$

⑥

There are gaps

- between 1 and 2: $f(n)$ is smaller than $n^{\log_b a}$, but not polynomially smaller
- between 2 and 3: $f(n)$ is larger than $n^{\log_b a}$, but not polynomially larger
- in case 3, if the regularity fails to hold

{Examples}:-

- ① $T(n) = 9T(n/3) + n$;
 — $a=9, b=3, f(n)=n$
 — $n^{\log_b a} = n^{\log_3 9} = \Theta(n^2)$
 — $f(n) = O(n^{\log_3 9 - \epsilon})$ for $\epsilon = 1$
 by case 1, $T(n) = \Theta(n^2)$
- ② $T(n) = T(2n/3) + 1$
 — $a=1, b=3/2, f(n)=1$
 — $n^{\log_b a} = n^{\log_{3/2} 1} = \Theta(n^0) = \Theta(1)$
 by case 2 $T(n) = \Theta(\lg n)$

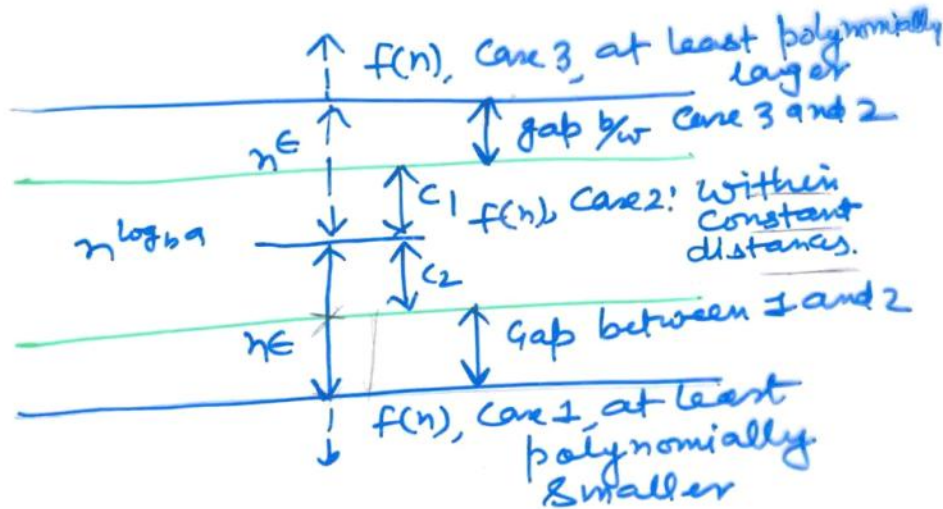
⑦

③ $T(n) = 3T(n/4) + n \lg n$
 — $a=3, b=4, f(n) = n \lg n$
 — $n^{\log_b a} = n^{\log_4 3} = \Theta(n^{0.793})$
 — $f(n) = \Omega(n^{\log_4 3 + \epsilon})$ for $\epsilon \approx 0.2$
 — Master case 3 for large n , the regularity holds for $C = 3/4$
 $a f(n/4) = 3(n/4) \lg(n/4) \leq (3/4) n \lg n = C f(n)$
 by case 3 $T(n) = \Theta(f(n)) = \Theta(n \lg n)$ ✓

④ $T(n) = 2T(n/2) + n \lg n$
 — $a=2, b=2, f(n) = n \lg n$
 — $n^{\log_b a} = n^{\log_2 2} = \Theta(n)$
 — $f(n)$ is asymptotically larger than $n^{\log_b a}$, but not polynomially larger because $f(n)/n^{\log_b a} = \lg n$, which is asymptotically less than n^ϵ for any $\epsilon > 0$

⑤ $T(n) = 9T(\lfloor n/3 \rfloor) + n^2 / \lg n$ | $T(n) = T(n/2) + n(\sin(n - \frac{\pi}{2}) + 2)$

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Note:-

- 1: for Case 3, the regularity also must hold
- 2: if $f(n)$ is $\lg n$ smaller, then fall in gap in 1 and 2
- 3: if $f(n)$ is $\lg n$ larger, then fall in gap in 3 and 2

